

# Electricity applications: heating, power, energy and exam preparation

## A1. Scope and source map

These original Class 10 notes extend the foundations note through applications, assumptions and interpretation.

Source: NCERT Science, Chapter 11, Electricity, Reprint 2026–27. Section 11.7 begins on printed page 188; its derivation continues on 189; section 11.7.1 occupies page 190. Section 11.8 actually starts on page 191 and continues onto 192, not 190. All page references below are printed pages. Official source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — sections 11.7–11.8, pp. 188–192.

All worked examples, experimental datasets, self-test questions and revision plans here are original teaching material, not official questions or marking schemes. Suggested priorities are not predictions: there are no invented prediction probabilities and no guarantee about examination content or marks.

## A2. Derive Joule’s heating law, not just its final line

Potential difference is work done per unit charge, so transferring charge  $Q$  through voltage  $V$  requires electrical energy  $W = VQ$ . For a steady current,  $Q = It$ . Substitution gives  $W = VIt$ . In a purely resistive device, electrical energy becomes heat, hence  $H = VIt$ . Using  $V = IR$  gives  $H = (IR)It = I^2Rt$ .

Thus heat depends on the square of current, resistance and duration. State the quantities held constant whenever describing proportionality. Doubling current gives four times the heat only when  $R$  and  $t$  remain unchanged. Doubling time doubles heat only when the electrical operating conditions remain unchanged.

Use amperes, ohms and seconds to obtain joules. Here  $W$  denotes work in the derivation; the unit symbol  $W$  elsewhere means watt. Do not assume that all input energy becomes useful heating in every appliance. Electrical energy may also produce motion or light; even in a heater, some heat escapes to the surroundings.

Source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.7, pp. 188–189, equations 11.19–11.21.

## A3. Fixed current versus fixed voltage: the decisive comparison

The statement “more resistance produces more heat” is incomplete. First identify what the source or connection keeps unchanged. Substituting  $I = V/R$  into  $H = I^2Rt$  also gives  $H = V^2t/R$ . These are equivalent expressions, not conflicting laws.

For a cord and heating element carrying the same current, the element’s greater resistance produces more electrical heating. Whether a part glows also depends on its temperature, dimensions and heat loss; resistance alone is not a complete temperature argument.

Source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.7, pp. 188–190. The fixed-voltage comparison is an algebraic consequence of the cited equations.

- Fixed  $I$  and  $t$ :  $H = I^2Rt$ , so a larger resistance produces more heat. Comparing parts carrying the same current uses this condition.
- Fixed  $V$  and  $t$ :  $H = V^2t/R$ , so a larger resistance produces less heat because the current decreases.
- Fixed  $R$  and  $t$ : doubling  $V$  doubles  $I$  and makes heat four times larger.

- If temperature changes  $R$  appreciably, a calculation using one unchanged resistance is only an approximation; see A8.

## A4. Useful heating and material choice

An iron transfers heat to fabric; a toaster heats food; a kettle heats water. Their purpose makes electrical heating useful rather than merely an unwanted loss. Heating elements commonly use alloys with relatively high resistivity that do not oxidise readily at high temperatures. Avoid saying that every high-resistance material is suitable: the element must also tolerate operating conditions.

An incandescent filament must become hot enough to emit light without melting. NCERT identifies tungsten's high melting point and describes an inactive gas filling that helps prolong filament life. Much of the input power appears as heat rather than visible light. This explanation concerns filament lamps, not every type of lamp.

Source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.5, p. 179; section 11.7.1, p. 190.

## A5. Power and appliance ratings

Power measures the rate of energy transfer:  $P = E/t$ . For electrical input,  $P = VI$ . For a resistor, substitution gives  $P = I^2R = V^2/R$ . Power is measured in watts;  $1 \text{ W} = 1 \text{ J/s}$  and  $1 \text{ kW} = 1000 \text{ W}$ . Heat and power are different: a modest power maintained for a long time can deliver substantial energy.

A rating such as “240 V, 1200 W” specifies operation at the stated voltage. It gives rated current  $I = P/V$  and operating resistance  $R = V^2/P$  for the resistive model. It does not mean that the device consumes 1200 joules in total, nor that its power remains 1200 W at every voltage.

At another voltage, use  $P = V^2/R$  only if resistance is assumed unchanged. Real operating resistance can depend on temperature. Nameplate power also does not establish how many hours an appliance is actively drawing that power.

Source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.7, p. 189; section 11.8, p. 191.

## A6. Energy, meter units and cost

Choose the unit route before substituting. Power in watts multiplied by time in seconds gives joules. Power in kilowatts multiplied by time in hours gives kilowatt-hours. A kilowatt-hour is energy, not power, and is commonly called one electricity “unit”.

Unless stated otherwise, a classroom cost problem assumes a constant tariff and omits taxes, standing charges and tariff slabs. A cycling appliance may draw its rated power only during part of the elapsed time. Do not invent an operating fraction when none is supplied.

Source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.8, pp. 191–192.

- $1 \text{ Wh} = 3600 \text{ J}$ ;  $1 \text{ kWh} = 1000 \text{ Wh} = 3.6 \times 10^6 \text{ J}$ .
- Joules to kWh: divide by  $3.6 \times 10^6$ . kWh to joules: multiply by  $3.6 \times 10^6$ .
- Energy charge = energy in kWh  $\times$  tariff per kWh. Keep currency units in the calculation.
- For several appliances, calculate each energy contribution and add them. Use actual operating time, not simply time plugged in.

## A7. Eight checked original worked examples

These examples assume steady conditions unless a change is explicitly described. The amounts and tariffs are invented for practice, not taken from official questions. Formula source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.7, pp. 188–189; section 11.8, pp. 191–192.

- Example 1 — timed heating: A  $12\ \Omega$  resistor carries  $1.5\ \text{A}$  for 4 minutes. Convert  $t = 240\ \text{s}$ .  $H = I^2Rt = 1.5^2 \times 12 \times 240 = 6480\ \text{J}$ . Check:  $P = 27\ \text{W}$ , and  $27 \times 240 = 6480\ \text{J}$ . Leaving time as 4 would understate heat sixtyfold.
- Example 2 — same resistors, different constraints: Compare  $5\ \Omega$  and  $10\ \Omega$  for 30 s. At a fixed  $2\ \text{A}$ , their heats are  $600\ \text{J}$  and  $1200\ \text{J}$ . Connected separately across  $10\ \text{V}$ , their heats are  $600\ \text{J}$  and  $300\ \text{J}$ . The ordering reverses because the second comparison fixes voltage, not current.
- Example 3 — reading a rating: An ideal heater is rated  $240\ \text{V}$ ,  $1200\ \text{W}$ .  $I = 1200/240 = 5\ \text{A}$ ;  $R = 240^2/1200 = 48\ \Omega$ . At  $120\ \text{V}$  with unchanged  $R$ ,  $P = 120^2/48 = 300\ \text{W}$  and  $I = 2.5\ \text{A}$ . Halving voltage quarters power under this assumption.
- Example 4 — kettle operating cost: A  $1.8\ \text{kW}$  kettle runs for 8 minutes daily for 25 days. Total time =  $200/60\ \text{h}$ .  $E = 1.8 \times 200/60 = 6\ \text{kWh} = 21.6 \times 10^6\ \text{J}$ . At an assumed ₹7.50/kWh, energy charge = ₹45. No additional bill charges are included.
- Example 5 — a small mixed-use bill: Four  $9\ \text{W}$  lamps operate 5 h/day, and a  $60\ \text{W}$  fan operates 6 h/day, for 30 days. Lamps use  $0.036 \times 5 \times 30 = 5.4\ \text{kWh}$ ; fan uses  $0.060 \times 6 \times 30 = 10.8\ \text{kWh}$ . Total =  $16.2\ \text{kWh}$ ; at ₹6/kWh, charge = ₹97.20.
- Example 6 — reverse calculation from heat: A  $16\ \Omega$  resistor produces  $7200\ \text{J}$  in 5 minutes.  $I^2 = H/(Rt) = 7200/(16 \times 300) = 1.5\ \text{A}^2$ . Current magnitude  $I = \sqrt{1.5} \approx 1.225\ \text{A}$ ;  $V = IR \approx 19.60\ \text{V}$ . Check using unrounded values:  $P = 24\ \text{W}$  and  $Pt = 7200\ \text{J}$ .
- Example 7 — energy from transferred charge: During a steady transfer,  $360\ \text{C}$  passes through a resistor across  $24\ \text{V}$  in 3 minutes.  $H = VQ = 8640\ \text{J}$ . Independently,  $I = 360/180 = 2\ \text{A}$ ;  $P = VI = 48\ \text{W}$ ;  $H = 48 \times 180 = 8640\ \text{J}$ . Charge is not measured in joules.
- Example 8 — elapsed time versus active time: A thermostatically controlled  $1.5\ \text{kW}$  heater is on for 40% of a 5 h period. Given that fraction, active time = 2 h and  $E = 3\ \text{kWh} = 10.8\ \text{MJ}$ . At ₹8/kWh, the energy charge is ₹24. Assuming uninterrupted heating would incorrectly give 7.5 kWh.

## A8. Experiment analysis: ideal relations and real temperature effects

Use only a teacher-supervised low-voltage heating setup, never a mains experiment. Measure current through the element and voltage across it; record heating time and the temperature change of a fixed amount of water. Keep the container, initial temperature, immersion depth and stirring procedure comparable. Let equipment cool before handling or repeating a run.

An ideal comparison might use  $R = 10\ \Omega$  and  $t = 60\ \text{s}$ . At  $0.50\ \text{A}$ , calculated electrical heat is  $150\ \text{J}$ ; at  $1.00\ \text{A}$ , it is  $600\ \text{J}$ . These are illustrative predictions, not measured classroom results. Doubling current predicts four times the electrical heat, but does not automatically prove a fourfold thermometer rise.

Some energy warms the container and element; some escapes to air. Longer or hotter runs can lose different amounts. Uneven stirring, delayed thermometer readings and instrument resolution also affect observations. Repeat trials, compare their spread and report disagreement rather than altering readings to match theory.

Temperature can change resistance itself. For an illustrative fixed  $6\ \text{V}$  supply, readings of  $0.60\ \text{A}$  initially and  $0.50\ \text{A}$  later imply  $R$  changing from  $10\ \Omega$  to  $12\ \Omega$  and power falling from  $3.6\ \text{W}$  to  $3.0\ \text{W}$ . These are hypothetical readings, not reported measurements. A cold resistance would

therefore overestimate sustained power in this example; do not generalise its direction to every material.

Record  $V$  and  $I$  during each run; for substantial changes, estimate energy over short intervals using each interval's  $VI$  and duration, then add. No temperature-coefficient formula is needed. NCERT explicitly limits the constant-resistance relation to unchanged temperature and notes that resistance varies with temperature; it supplies no quantitative temperature model here.

Source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.4, p. 176; section 11.5, p. 179; section 11.7.1, p. 190. The protocol and interpretation above are original guidance.

## A9. Essential formula revision with section anchors

Official source for these revision anchors: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — exact sections and printed pages are given in each bullet.

- A2:  $W = VQ$ ;  $Q = It$ ; resistive heat  $H = VIt = I^2Rt$ . NCERT section 11.7, pp. 188–189.
- A3:  $H = V^2t/R$  for a resistor; identify fixed  $I$  or fixed  $V$  before comparing. NCERT section 11.7, pp. 188–189, with algebraic substitution.
- A5:  $P = E/t = VI$ ; resistive forms  $P = I^2R = V^2/R$ . Rating relations  $I = P/V$  and  $R = V^2/P$  follow by rearrangement. NCERT section 11.8, p. 191.
- A6:  $E = Pt$ ;  $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$ ; energy charge =  $\text{kWh} \times \text{tariff}$ . NCERT section 11.8, pp. 191–192.
- A8: constant- $R$  calculations require attention to temperature. NCERT section 11.4, p. 176; section 11.5, p. 179.

## A10. Misconceptions and quick repairs

Source: <https://ncert.nic.in/textbook/pdf/jesc111.pdf> — section 11.7, pp. 188–190; section 11.8, p. 191. These repairs apply the source relations rather than reproduce official answers.

- “Higher wattage always uses more energy.” Compare power multiplied by operating time, not power alone.
- “kW/h and kWh mean the same.” Write kWh for energy: multiplication by hours, not division.
- “Higher resistance always heats more.” Name the controlled quantity; return to A3.
- “A watt is a joule.” A watt is a joule per second.
- “Electrons are used up.” Energy is transferred; electrons are not consumed as fuel.
- “The calculated resistor heat all enters the water.” Separate electrical input from the portion reaching the intended object.

## A11. Exam method and partial-mark guidance

Read the required quantity first. List known values with units; convert minutes to seconds for joules, or to hours for kWh. Select a relation containing the unknown and available data. Write the symbolic equation before substitution, show one clear calculation line, and finish with the quantity and unit.

For comparisons, state the common voltage or current before the ratio. For explanations, connect condition, equation and conclusion. For derivations, retain the steps  $VQ$ ,  $Q = It$ ,  $H = VIt$  and  $V = IR$  instead of jumping directly to the result.

Clear working makes correct intermediate reasoning visible and may support partial credit even after an arithmetic slip. This is study guidance, not an official rubric: marks depend on the question and its marking scheme. If stuck, write a relevant relation and unit conversion, then

return later. Do not manufacture missing data. Finally check whether the numerical size and direction of change are plausible.

## A12. A realistic one-week revision plan

Allow about 40 minutes daily, with a short break when needed. Begin each session with five minutes of closed-book recall and finish by recording one error and its correction. Adapt the workload to your timetable rather than sacrificing sleep.

During the final 24 hours, use one short session for formulas and one for your error log. Recheck fixed-current versus fixed-voltage reasoning and time conversions. Avoid a new collection of speculative questions. Prepare required materials, stop heavy study early and protect normal sleep; a brief final recall is enough.

- Day 1: Read A2 and reconstruct the derivation twice without copying. Explain every symbol aloud.
- Day 2: Study A3–A4; redo Example 2 and explain why its ordering reverses.
- Day 3: Practise ratings using A5 and Examples 1, 3 and 6. Check units and assumptions.
- Day 4: Use A6 and Examples 4, 5, 7 and 8. Practise both energy-unit routes.
- Day 5: Analyse A8. List controlled variables, likely errors and what changing  $V/I$  reveals.
- Day 6: Attempt A13 closed-book in 15 minutes; use remaining time to repair weak reasoning.
- Day 7: Revisit errors, reproduce A9 from memory and complete two previously missed examples unaided.

## A13. Original self-test and answers

Try these without looking back. Award yourself completion only when you can explain the condition or unit, not merely recognise a number.

Review A3 for comparisons, A5 for ratings, A6 for conversions and A8 for experiments.

- Test 1: Current through an unchanged resistor becomes three times larger for the same duration. What happens to heat? Answer: nine times larger, because  $H$  depends on  $I^2$  at fixed  $R$  and  $t$ .
- Test 2: At fixed voltage and duration, resistance doubles. Answer: heat halves; current also halves.
- Test 3: A 75 W device operates for 2 h. Answer:  $150 \text{ Wh} = 0.15 \text{ kWh} = 540000 \text{ J}$ .
- Test 4: A 200 V, 800 W ideal heater operates at its rating. Answer:  $I = 4 \text{ A}$ ;  $R = 50 \Omega$ .
- Test 5: Convert  $9 \times 10^6 \text{ J}$  to units and calculate the energy charge at ₹6.40/unit. Answer: 2.5 kWh and ₹16.
- Test 6: At unchanged voltage, current decreases during warming. What should you check? Answer: calculate  $V/I$  at each reading; an increase indicates increased resistance. Check source stability and contacts before attributing the change entirely to temperature.

## Official curriculum scope and assessment limitation

CBSE Science (086), Class X, 2026–27, PDF p.6 Unit IV lists electric current, potential difference, resistance, resistivity, Ohm's law, resistor combinations, heating and power. Practical 4 on PDF p.7 covers  $V$ - $I$  measurements; practical 5 on PDF p.8 covers series and parallel equivalent resistance. Source: [https://cbseacademic.nic.in/web\\_material/CurriculumMain27/SecPart1/Science\\_SecP1\\_2026-27.pdf](https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Science_SecP1_2026-27.pdf)

The same PDF p.6 contains an unresolved teacher-note wording inconsistency naming “Electric Effects of Electric Current” among unassessed topics, despite the explicit Unit IV list. These are chapter study materials, not a guarantee of summative assessment coverage; consult school/CBSE clarification. Unit IV’s 13 marks covers a combined unit, not an Electricity-only allotment.